

Marginal Rate of Substitution Between Public and Private Goods

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Representative household : derives utility from public & private goods & lab

Utility function:

$$U(c, g)$$

share of production, private workers

share of production, public workers

- U is strictly increasing in c
- U is strictly increasing in g
- U is strictly concave

Marginal rate of substitution b/w public & private goods

$$MRS_{g,c} = \frac{\partial U / \partial g}{\partial U / \partial c} > 0$$

- $MRS_{g,c}$ is strictly decreasing in g/c
- $MRS_{g,c}(g/c = 0) > 1$ $\rightarrow$ so that it is desirable to provide some public goods (interim solution to planner's problem)